

- Consider the following simple model of health insurance. Let there be three types of people, LOW, MEDIUM, and HIGH risk, whose yearly medical expenses are \$0 with 95% probability and \$25K, \$50K, and \$100K (respectively) with 5% probability. All three risk types are equally represented in the population, and all people have an initial wealth level of \$100K, and a utility function over wealth of $u(x) = \sqrt{x}$ (where x is measured in thousands of dollars).

Insurers only offer the simplest policy, where people pay a premium but no deductible and medical expenditure is fully reimbursed. Assume that insurers have no costs other than the expected payouts to their policyholders.

- Please fill in the following table:

Customer risk type	Expected cost of customer type to insurer	Highest premium consumer would pay
LOW		
MEDIUM		
HIGH		

- If it were required that all people be insured, and insurance is competitive so that premiums equal average cost, what would the premium be?
- If insurance were optional, what premium would insurers charge? (Note: they would consider charging one of the three premiums solved for in part a. If they charged the lowest, all three types would buy; if they charge the 2nd lowest, two types would buy; if they charged the highest, only one type would buy). Again, insurance is competitive, so that premiums equal average cost.

And which risk types would choose to buy insurance at that premium?

CIRCLE THOSE THAT APPLY:

LOW MEDIUM HIGH NONE

SOLUTION:

2a) The highest premium a person is willing to pay, p (where l is the loss in the bad state of the world) is the solution to $5\% \times \sqrt{100 - l} + 95\% \times \sqrt{100} = \sqrt{100 - p}$.

Expected cost to the insurer is $5\% \times l$. Filling out the table:

Customer risk type	Expected cost to insurer	Highest premium consumer would pay
LOW	\$1250	\$1335
MEDIUM	\$2500	\$2907

HIGH	\$5000	\$9750
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2b) The average cost is $(1/3)*(\$1250+\$2500+\$5000)=\2917 . This is the actuarially fair premium.

2c) If the insurer prices at \$1335 or below, its average cost will be

$(1/3)*(\$1250+\$2500+\$5000)=\2917 . Negative profit.

If the insurer prices between \$1336 and \$2907, its average cost will be

$(1/2)*(\$2500+\$5000)=\$3750$. Negative profit.

If the insurer prices between \$2908 and \$9750, its average cost will be \$5000.

So it can avoid negative profit if it serves HIGH types only, at an actuarially fair premium of \$5000.